

Part 1

Thursday, October 19, 2023 10:41 AM

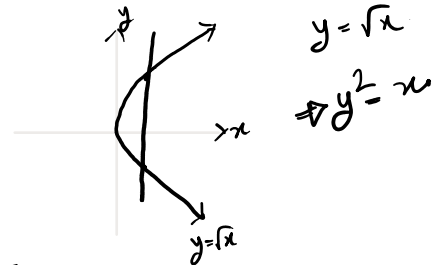
Differential Calculus

Function: A function is a rule which assigns for each independent variable x , there is exactly one dependent variable y .

We write: $y=f(x)$ to mean that (x,y) is a $y=\sqrt{x}$ point on the graph of f

Here, $y=\sqrt{x} \rightarrow$ represents a function

$y^2=x \rightarrow$ does not represent a function

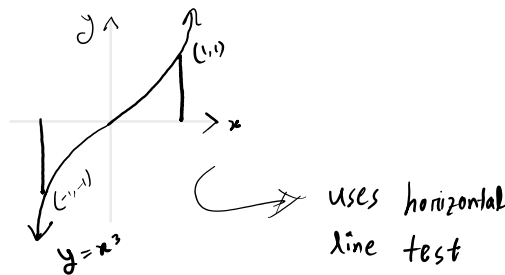


Ex: $y = \sqrt{1-x^2}$ (semi-circle)

$\hookrightarrow$ uses vertical line test

One to One Function $\left\{ \begin{matrix} a \neq b \\ f(a) \neq f(b) \end{matrix} \right\}$

$f(x) = x^3 \rightarrow$ one to one function



Families of functions (Library of Functions)

1 Constant Function:

$$\text{Dom}(f) = \mathbb{R} = (-\infty, +\infty)$$

$$\text{Range}(f) = \{b\}$$

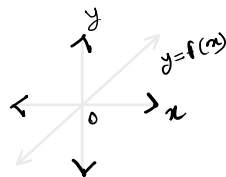


2 Identity Function:

$$f(x) = x$$

$$\text{Dom}(f) = \mathbb{R}$$

$$\text{Range}(f) = \mathbb{R}$$



3 Square Function:

$$y = f(x) = x^2$$

$$\text{Dom}(f) = \mathbb{R}$$

$$\text{Range}(f) = [0, \infty)$$

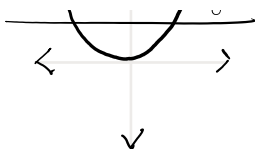


$$\text{Dom}(f) = \mathbb{R}$$

$$\text{Range}(f) = [0, +\infty)$$

Also, an even function.

$$f(-x) = f(x)$$



symmetric with respect to the y-axis

4] Cube Function:

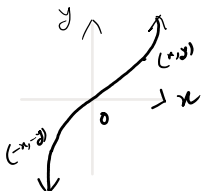
$$(1,1) \in f$$

$$(-1,-1) \in f$$

$$n \in \text{Dom } f$$

$$-n \in \text{Dom } f$$

$$f(-n) = -f(n)$$



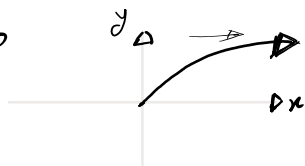
Also an odd function

$$f(-x) = -f(x)$$

symmetric with respect to the origin.

5] Square root function:

$$f(x) = \sqrt{x}, \quad x \geq 0$$



x-intercept is 0

y-intercept is 0

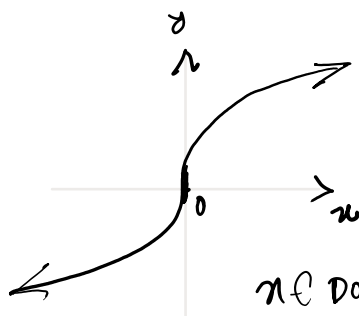
$$\text{Dom}(f) = [0, +\infty)$$

$$\text{Range}(f) = [0, +\infty)$$

6] Cube root function

$$y = f(x) = \sqrt[3]{x}$$

$$f(-x) = \sqrt[3]{-x}$$



$$x \in \text{Dom}(f)$$

$$-x \in \text{Dom}(f)$$

$$\text{Dom}(f) = \mathbb{R}$$

$$\text{Range}(f) = \mathbb{R}$$

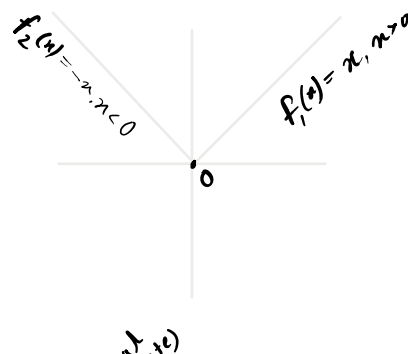
7] Absolute Value Function:

$$f(x) = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

(piece-wise defined function)

$$\text{Dom}(f) = \mathbb{R}$$

$$\text{Range}(f) = [0, +\infty)$$



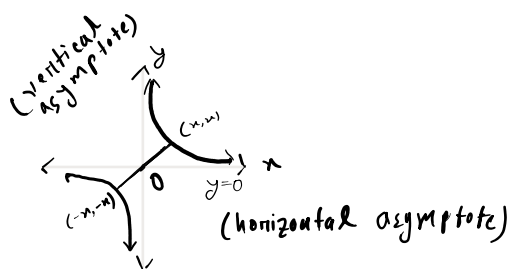
8] Reciprocal Function:

81 Reciprocal Function:

$$y = f(x) = \frac{1}{x} = \frac{p(x)}{q(x)}$$

$$f(-x) = -\frac{1}{x} = -f(x)$$

$\therefore f$ is odd.



Given $f(x) = \frac{\sqrt{x+3}}{x-2}$. Find $\text{dom}(f)$.

i) $\sqrt{x+3}$ is defined if $x+3 \geq 0 \Rightarrow x \geq -3$

ii) $x-2 \neq 0 \Rightarrow x \neq 2$

$$\text{Dom}(f) = [-3, +\infty) \setminus \{2\} = [-3, 2) \cup (2, +\infty)$$

Graphing techniques using transformations.

Composition of functions:
(New functions from old)

$$(f \circ g)(x) = f(g(x))$$

Ex: Given $f(x) = x^2$
 $g(x) = f(x) + k, \quad k > 0$
 $g(x) = x^2 + 3.$

$$\left| \begin{array}{l} y = f(x) = x^2 \\ (0,0), (1,1), (-1,1), (2,4), (-2,4) \\ \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ (0,3), (1,4), (-1,4), (2,7), (-2,7) \end{array} \right.$$

Ex: $f(x) = x^2 + 3$
 $g(x) = \sqrt{x}$

$$\therefore \text{Dom}(f) = \mathbb{R}; \quad \text{Dom}(g) = [0, +\infty)$$

$$(f \circ g)(x) = f(g(x)) = f(\sqrt{x}) = (\sqrt{x})^2 + 3 = x + 3$$

$$\therefore \text{Dom}(f \circ g) = [0, +\infty)$$

$$\hookrightarrow (g \circ f)(x) = g(x^2 + 3) = \sqrt{x^2 + 3}$$

$$\therefore \text{Dom}(g \circ f) = \mathbb{R}$$

Ex: $f(x) = \frac{x}{x+2}, \quad g(x) = \frac{1}{x+3}$

$$\text{Dom}(f) = \mathbb{R} \setminus \{-2\}, \quad \text{Dom}(g) = \mathbb{R} \setminus \{-3\}$$

$$\text{Now, } (f \circ g)(x) = f(g(x))$$

$$= f\left(\frac{1}{x+3}\right)$$

$$= \frac{\frac{1}{x+3}}{\frac{1}{x+3} + 2}$$

$$= \frac{1}{2x+7}.$$

Ex: $f(x) = 1 + \sqrt{x+2}$
 $g(x) = x-3$

$$(f+g)(x) = f(x) + g(x) = 1 + \sqrt{x+2} + x-3 = x-2 + \sqrt{x+2}$$

$$(f-g)(x) = f(x) - g(x) = 1 + \sqrt{x+2} - x + 3 = 4 - x + \sqrt{x+2}$$

$$(fg)(x) = f(x)g(x) = (1 + \sqrt{x+2})(x-3) = x-3 + (x-3)\sqrt{x+2}$$

$$\therefore \text{Dom}(f+g) = \text{Dom}(f-g) = \text{Dom}(fg) = [-2, +\infty)$$

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{1 + \sqrt{x+2}}{x-3}$$

$$\therefore \text{Dom}\left(\frac{f}{g}\right) = [-2, +\infty) \setminus \{3\}$$

Ex:

i) $f(x) = \frac{1}{(x-1)(x-3)}$

$$\text{Dom}(f) = \{x \in \mathbb{R} : x \neq 1 \text{ and } x \neq 3\}$$

$$= (-\infty, 1) \cup (1, 3) \cup (3, +\infty)$$

ii) $g(x) = \tan x = \frac{\sin x}{\cos x}$ is defined for all x

except where $\cos x = 0$.

$$\Rightarrow x = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \dots$$

iii) $h(x) = \sqrt{x^2 - 5x + 6}$
 $= \sqrt{(x-2)(x-3)}$

$$(x-2)(x-3) \geq 0$$

$$\Rightarrow (-\infty, 2] \cup [3, +\infty)$$

$$\text{Dom}(h) = (-\infty, 2] \cup [3, +\infty)$$

Transformations

i) $y = f(x) + k$ (vertical shift up/down)

$$k > 0 \rightarrow \text{up}$$

$$k < 0 \rightarrow \text{down}$$

ii) $y = f(x+c)$ (horizontal shift left/right)

$$c > 0 \rightarrow \text{left}$$

$$c < 0 \rightarrow \text{right}$$

$$f(x) = 2 + \sqrt{x-1}$$

i) $f_1(x) = \sqrt{x}$ (0,0), (4,2), (9,3)

ii) $f_2(x) = \sqrt{x-1}$ (1,0), (5,2), (10,3)

$\hookrightarrow$ Horizontal shift 1 unit right.

iii) $f_3(x) = 2 + \sqrt{x-1}$ (1,2), (5,4), (10,6)

$\hookrightarrow$ vertical shift 2 unit up.

iii) $y = +\sqrt{x} = f_1(x)$

$y = -\sqrt{x} = f_2(x)$

$$y = f(x) \rightarrow y = -f(x)$$

$\hookrightarrow$ Reflection about the x-axis.

$$y = f(x)$$

$\downarrow$

$$y = g(x) = f(-x)$$

$\hookrightarrow$ Reflection about the y-axis.

$$y = \sqrt[3]{2-x}$$

i) $y = \sqrt[3]{x}$

ii) $y = \sqrt[3]{-x}$ (reflection about the y-axis)

iii) $y = \sqrt[3]{-x+2}$ (horizontal shift 2 unit right)

$$= \sqrt[3]{-(x-2)}$$

⊗ Graph the function: $y = 4 - |x-2|$ using transformation

⊗ Graph the function : $y = 4 - |x-2|$ using transformation

Polynomial and Rational Functions

$$f(x) = p_n(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

$$a_n \neq 0$$

$a_n, a_{n-1}, \dots, a_1, a_0$ are real co-efficients

a_n : leading co-efficient of f

a_0 : constant co-efficient of f of degree n .

⊗ $f(x) = 0 \rightarrow$ zero polynomial but degree is undefined

⊗ $f(x) = \frac{\sqrt{x-2}}{x+3}$ is not a polynomial

⊗ $f(x) = \frac{x+2}{x^2+5}$ is a rational function

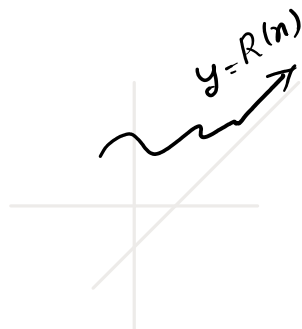
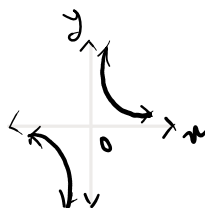
$$= \frac{p(x)}{q(x)} \quad \left| \begin{array}{l} p(x): a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 \\ q(x): b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x + b_0 \end{array} \right. \begin{array}{l} \text{i) If } n \geq m, f \text{ is proper} \\ \text{ii) If } m > n, f \text{ is improper} \end{array}$$

Reciprocal Function

$$f(x) = \frac{2}{x}$$

As $x \rightarrow -\infty$ or $x \rightarrow +\infty$

$$f(x) \rightarrow 0$$



Limit of Functions

$$\lim_{x \rightarrow c} f(x) = L$$

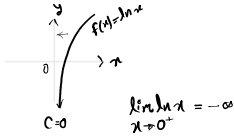
$$x \rightarrow c.$$

$$\Leftrightarrow f(x) \rightarrow L \text{ as } x \rightarrow c.$$

Right hand limit:

$$\lim_{x \rightarrow c^+} f(x) = L_1$$

(limit approaches
from right to left)



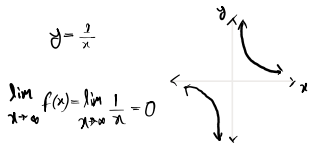
$$\lim_{x \rightarrow 0^+} \ln x = -\infty$$

$$\lim_{x \rightarrow c^-} f(x) = L_2$$

If $L_1 = L_2$, then we have the two-sided limit and

$$\text{we write, } \lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = L$$

$$y = \frac{1}{x}$$



$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

$$\lim_{x \rightarrow -\infty} f(x) = \lim_{x \rightarrow -\infty} \frac{1}{x} = 0$$

$$\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty \rightarrow \text{limit does not exist}$$

$$\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty \rightarrow \text{limit does not exist}$$

Evaluation of limits:

$$1) \lim_{x \rightarrow 3} (x^3 - 3x^2 + 9x) = 27$$

$$2) \lim_{x \rightarrow 0} \frac{6x - 9}{x^2 - 12x + 3} = \frac{0 - 9}{0^2 - 12 \cdot 0 + 3} = \frac{-9}{3} = -3$$

$$3) \lim_{x \rightarrow 2} \frac{x^3 + 8}{x + 2} = \frac{x^3 + 8}{x + 2} = x^2 - 2x + 4, x \neq -2$$

$$= 12$$

$$4) \lim_{x \rightarrow 3^-} \frac{x}{x-3} = \lim_{h \rightarrow 0} \frac{3-h}{3-h-3} = \lim_{h \rightarrow 0} \frac{3-h}{-h} = -\infty$$

$$5) \lim_{x \rightarrow 1} \frac{x}{x^2 - 4} = \lim_{h \rightarrow 0} \frac{2-h}{(2-h)^2 - 4}$$

$$= \lim_{h \rightarrow 0} \frac{2-h}{4 - 4h + h^2 - 4}$$

$$= \lim_{h \rightarrow 0} \frac{2-h}{h(h-4)}$$

$$= -\infty$$

Evaluation of limits: (main idea)

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x)$$

Ex 1:

$$\text{Given } f(t) = \begin{cases} t-2, & t < 0 \quad \leftarrow (-\infty, 0) \\ t^2, & 0 \leq t \leq 2 \quad \leftarrow [0, 2] \\ 2t, & t > 2 \quad \leftarrow (2, \infty) \end{cases}$$

Find (i) $\lim_{t \rightarrow 0^-} f(t)$, (ii) $\lim_{t \rightarrow 0^+} f(t)$, (iii) $\lim_{t \rightarrow 0} f(t)$, if they exist

$$2t, t > 2 \leftarrow \begin{matrix} 2, 2 \\ (2, \infty) \end{matrix}$$

Find (i) $\lim_{t \rightarrow 0} f(t)$, (ii) $\lim_{t \rightarrow 1} f(t)$, (iii) $\lim_{t \rightarrow 2} f(t)$, if they exist

Soln:

$$(i) \lim_{t \rightarrow 0^-} f(t) = \lim_{t \rightarrow 0^-} (t+2) = 0+2 = 2.$$

$$\lim_{t \rightarrow 0^+} f(t) = \lim_{t \rightarrow 0^+} (t^2) = 0^2 = 0$$

Since $\lim_{t \rightarrow 0^-} f(t) \neq \lim_{t \rightarrow 0^+} f(t)$, $\lim_{t \rightarrow 0} f(t)$ does not exist. $t \in [0, 2]$

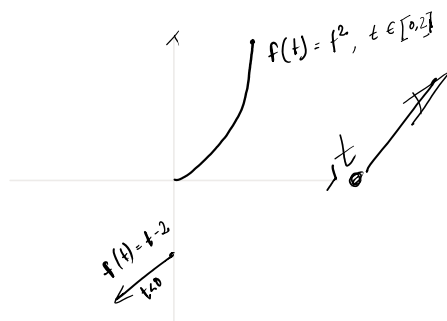
$$(ii) \lim_{t \rightarrow 1} f(t) = \lim_{t \rightarrow 1} t^2 = 1^2 = 1.$$

$$(iii) \lim_{t \rightarrow 2^-} f(t) = \lim_{t \rightarrow 2^-} (t^2) = 2^2 = 4$$

$$\lim_{t \rightarrow 2^+} f(t) = \lim_{t \rightarrow 2^+} (t^2) = 2^2 = 4$$

$$\text{Since } \lim_{t \rightarrow 2^-} f(t) = \lim_{t \rightarrow 2^+} f(t) = 4$$

$$\lim_{t \rightarrow 2} f(t) = 4.$$



H.W: $f(x) = \begin{cases} x-1, & x \leq 3 \\ 3x-7, & x > 3 \end{cases}$

Find $\lim_{x \rightarrow 3} f(x)$, if it exists

Theorems on Limits

$$\text{If } \lim_{x \rightarrow c} f(x) = L_1, \lim_{x \rightarrow c} g(x) = L_2,$$

$$\text{then, (i) } \lim_{x \rightarrow c} [f(x) \pm g(x)] = \lim_{x \rightarrow c} f(x) \pm \lim_{x \rightarrow c} g(x) = L_1 \pm L_2.$$

$$(ii) \lim_{x \rightarrow c} [f(x) \cdot g(x)] = \lim_{x \rightarrow c} f(x) \cdot \lim_{x \rightarrow c} g(x) = L_1 L_2$$

$$(iii) \lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)} = \frac{L_1}{L_2} \quad \text{if } g(x) \neq 0$$

Continuity

A function f is said to be continuous at $x=c$ if the following statements are satisfied:

(i) $f(c)$ is defined

statements are satisfied:

- i) $f(c)$ is defined
- ii) $\lim_{x \rightarrow c} f(x)$ exists
- iii) $\lim_{x \rightarrow c} f(x) = f(c)$

Theorem 1: If f and g are continuous at $x=c$, then

- i) $f \pm g$ is continuous at $x=c$
- ii) fg is continuous at $x=c$
- iii) $\frac{f}{g}$ is continuous at $x=c$ if $g(c) \neq 0$.

Theorem 2:

- i) A polynomial function is continuous everywhere.

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

$$0 = 0x^n + 0x^{n-1} + \dots + 0x + 0$$

- ii) A rational function is continuous everywhere except where the denominator is zero.

$$f(x) = \tan x$$

Ex: Given $f(x) = \begin{cases} -x^2, & x \leq 0 \\ 4x-4, & 0 < x \leq 1 \\ 4x^2-3x, & 1 < x < 2 \\ 3x+4, & x \geq 2 \end{cases}$

- i) Graph the function

- ii) Examine the continuity of

f at $x=0, 1, 2$

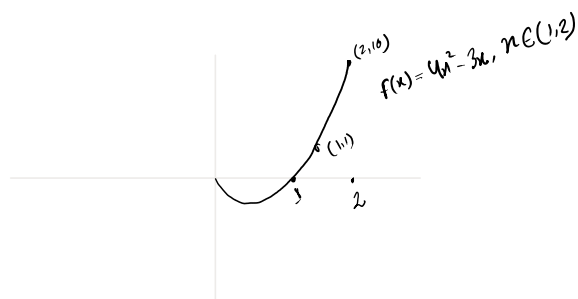
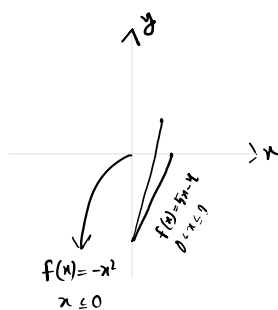
If, $f(x) = 4x^2 - 3x, x \in (1, 2)$

$$x = 0, \frac{3}{4}$$

$$y = 0$$

$$x = -\frac{-3}{2 \times 4} = \frac{3}{8}$$

$$f\left(\frac{3}{8}\right) = -\frac{9}{16}$$

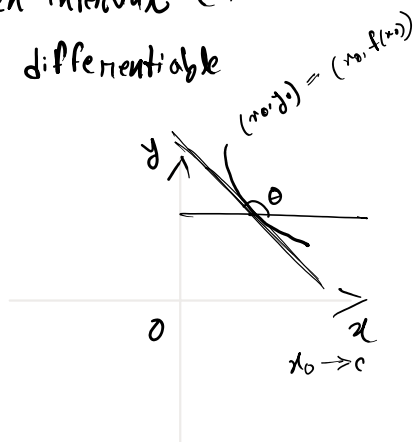


Differentiability

The function f' defined by $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ is known as the derivative of f with respect to x .

A function f is said to be differentiable at $x=c$ if $f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$ exists. If f is differentiable at each point in an open interval (a,b) then we say that f is differentiable on (a,b) .

Geometrically, $f'(c)$ represents the slope of a tangent line to the graph of f at $x=c$.



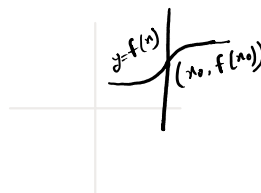
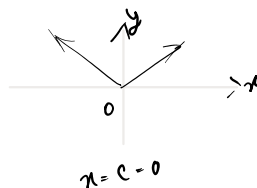
Theorem $\Rightarrow$ If f is differentiable at $x=c$, then f is continuous at that point.

In general, f is differentiable on an interval of the form $[a,b]$, $(-\infty, b]$, $[a, +\infty)$, $(a,b]$, $[a,b)$ if it is differentiable at all points inside the interval.

The process of finding a derivative is called differentiation.

A continuous function f at $x=c$ may fail to be differentiable at $x=c$ for the following reasons:

- i) $x=c$ is a corner point,
- ii) $x=c$ is a point of vertical tangency.



$$\left. \begin{aligned} \lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^-} (-x) = 0 \\ \lim_{x \rightarrow 0^+} f(x) &= \lim_{x \rightarrow 0^+} (x) = 0 \end{aligned} \right\} \therefore f \text{ is continuous at } x=0$$

$f'_L(c)$: Left hand derivative of f at $x=c$

$f'_R(c)$: Right hand derivative of f at $x=c$

$$f'_L(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{-h-0}{h} = -1$$

$$f'_R(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h-0}{h} = 1$$

Since $f'_L(0) \neq f'_R(0)$, f is not differentiable at $x=0$
 on (f does not exist at $x=0$)

Ex: $f(x) = \begin{cases} 1 + \sin x & x < \frac{\pi}{2} \\ 2 + (x - \frac{\pi}{2})^2 & x \geq \frac{\pi}{2} \end{cases}$; Discuss the existence of $f'(x)$ at $x = \frac{\pi}{2}$.

$$\begin{aligned} f'_L\left(\frac{\pi}{2}\right) &= \lim_{h \rightarrow 0} \frac{f\left(\frac{\pi}{2}+h\right) - f\left(\frac{\pi}{2}\right)}{h} \\ &= \lim_{h \rightarrow 0} \frac{1 + \sin\left(\frac{\pi}{2}+h\right) - 2}{h} \\ &= \lim_{h \rightarrow 0} \left(\frac{1}{h}\right) (-1 + \cos h) \\ &= \lim_{h \rightarrow 0} \left(\frac{1}{h}\right) \left[-1 + \left(-\frac{h^2}{2!} + \frac{h^4}{4!} - \dots\right)\right] \\ &= \lim_{h \rightarrow 0} \left(\frac{1}{h} \cdot h\right) \left[-\frac{h}{2!} + \frac{h^3}{4!} - \dots\right] \\ &= 0 \end{aligned}$$

$$f'_R\left(\frac{\pi}{2}\right) = \lim_{h \rightarrow 0} \frac{f\left(\frac{\pi}{2}+h\right) - f\left(\frac{\pi}{2}\right)}{h} = \lim_{h \rightarrow 0} \frac{2 + \left(\frac{\pi}{2}+h - \frac{\pi}{2}\right)^2 - 2}{h} = \lim_{h \rightarrow 0} \frac{h^2}{h} = 0$$

Since $f'_L\left(\frac{\pi}{2}\right) = f'_R\left(\frac{\pi}{2}\right)$, $f'(x)$ exists at $x = \frac{\pi}{2}$.

(f is differentiable at $x = \frac{\pi}{2}$)

Ex:

Find an eqn of the tangent line to the curve $f(x) = x - \frac{2}{x}$ at $(2,1)$ on this curve.

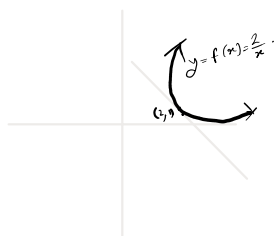
Find an eqn of the tangent line to the curve $f(x) = y = \frac{2}{x}$ at $(2, 1)$ on this curve.

Soln:
 $y = \frac{1}{x} = f(x)$

$(1, 1), (2, \frac{1}{2})$

$y = \frac{2}{x}$

$(1, 2), (2, 1)$



$$m_{\text{tan}} = f'(2) = \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0} \frac{\frac{2}{2+h} - 1}{h} = - \lim_{h \rightarrow 0} \frac{h}{h(2+h)} = - \lim_{h \rightarrow 0} \frac{1}{2+h} = -\frac{1}{2}$$

$\therefore$ The eqn of the line is $y - 1 = -\frac{1}{2}(x - 2) \Rightarrow y = -\frac{1}{2}x + 2$

Ex: Find the slopes of tangent lines to the graph of $y = \sqrt{x}$ at $x_0 = 1, x_0 = 4, x_0 = 9$

Soln:

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0+h) - f(x_0)}{h} = \frac{1}{2\sqrt{x_0}}$$

$f'(1) = \frac{1}{2}, f'(4) = \frac{1}{4}, f'(9) = \frac{1}{6}$

Logarithmic differentiation is useful for differentiating functions that are composed of products, quotients and powers.

Ex: $y = f(x) = \frac{x^2 \sqrt[3]{7x-14}}{(1+x)^4}$. Find $\frac{dy}{dx}$.

Soln:

$$y = \frac{x^2 \sqrt[3]{7x-14}}{(1+x)^4}$$

$$\hookrightarrow \ln y = \ln \frac{x^2 \sqrt[3]{7x-14}}{(1+x)^4}$$

$$= \ln(x^2) + \ln(\sqrt[3]{7x-14}) - \ln((1+x)^4)$$

$$= 2 \ln x + \frac{1}{3} \ln(7x-14) - 4 \ln(1+x)$$

$y = \ln$

$$\frac{d}{dx} (\ln y) = \frac{d}{dy} (\ln y) \frac{dy}{dx} = \frac{1}{y} \cdot y' = \frac{y'}{y}$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{2}{x} + \frac{1}{3} \cdot \frac{7}{7x-14} - 4 \cdot \frac{1}{1+x}$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{2}{x} + \frac{1}{3} \cdot \frac{7}{7x-14} - 4 \cdot \frac{4x}{1+x^2}$$

$$= \frac{2}{x} + \frac{7/3}{7x-14} - \frac{8x}{1+x^2}$$

$$\therefore \frac{dy}{dx} = \frac{x^{23} \sqrt{7x-14}}{(1+x^2)^4} \left[\frac{2}{x} + \frac{7/3}{7x-14} - \frac{8x}{1+x^2} \right]$$

Ex 2: Find $\frac{dy}{dx}$, where $y = \ln((\cos^2 x) \sqrt{1+x^4})$

Ex 3: u , where $y = \frac{(x^2-8)^3 \sqrt{x^3+1}}{x^6-7x+5}$

Ex 4: u , where $y = \frac{\sin x \cos x \tan^2 x}{\sqrt{x}}$

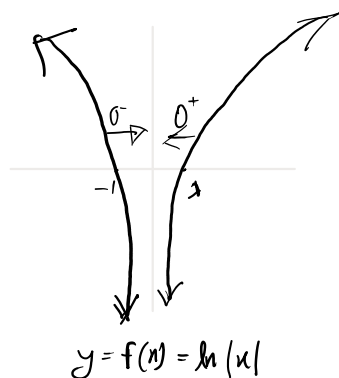
$$y = \ln |x|$$

i) $x > 0 \Rightarrow y = \ln x$

$$\frac{dy}{dx} = \frac{1}{x}$$

ii) $x < 0 : |x| = -x$

$$\frac{dy}{dx} = -\frac{1}{x}$$



Ex 5: $y = [(x^2+1)^{\sin x}]$, Find $\frac{dy}{dx}$.

Soln: $y = [(x^2+1)^{\sin x}]$

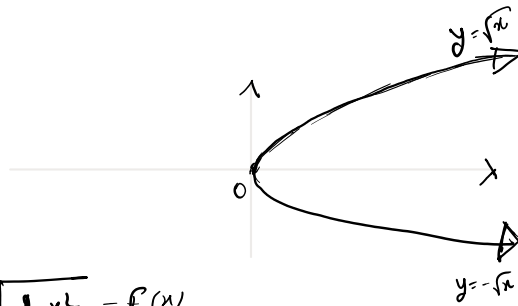
$$\Rightarrow \ln y = \sin x \ln(x^2+1)$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \cos x \ln(x^2+1) + \frac{2x \sin x}{x^2+1}$$

$$\Rightarrow \frac{dy}{dx} = \left\{ \ln(x^2+1) \cos x + \frac{2x \sin x}{x^2+1} \right\} (x^2+1)^{\sin x}$$

Implicit Differentiation

A given function in x, y defines the function f implicitly if the graph of $y = f(x)$ coincides with a portion of the graph of the eqn.



For example: i) $x^2 + y^2 = 1$ defines f implicitly $y = \pm \sqrt{1-x^2} = f(x)$
 $f_1(x) = \sqrt{1-x^2}$, $f_2(x) = -\sqrt{1-x^2}$

ii) $x = y^2$ implicitly defines the functions $f_1(x) = \sqrt{x}$ and $f_2(x) = -\sqrt{x}$.

Ex 1: Use implicit differentiation to find $\frac{dy}{dx}$, where $\cos y^2 + \sin y = x^2$

Soln:

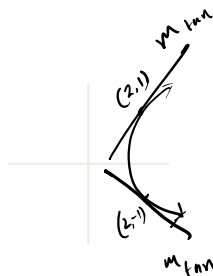
$$\begin{aligned} \cos y^2 + \sin y &= x^2 \\ \Rightarrow 10y \frac{dy}{dx} + \cos y \frac{dy}{dx} &= 2x \\ \Rightarrow \frac{dy}{dx} &= \frac{2x}{10y + \cos y} \end{aligned}$$

Ex 2: Find the slopes of the tangent lines to the graph of $y^2 - x + 1 = 0$ at $(2, -1)$ and $(2, 1)$

Soln:

$$\begin{aligned} y^2 - x + 1 &= 0 \\ \Rightarrow 2y \frac{dy}{dx} - 1 &= 0 \\ \Rightarrow \frac{dy}{dx} &= \frac{1}{2y} \end{aligned}$$

$$\therefore \left. \frac{dy}{dx} \right|_{(2,1)} = \frac{1}{2 \times 1} = \frac{1}{2}, \quad \left. \frac{dy}{dx} \right|_{(2,-1)} = -\frac{1}{2}$$

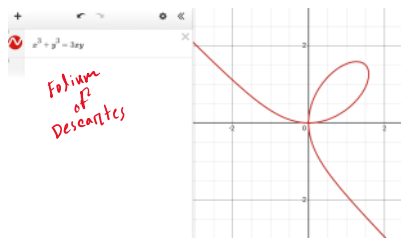


⊗ Folium of Descartes

$$\hookrightarrow x^3 + y^3 = 3xy$$

$$\Rightarrow 3x^2 + 3y^2 \frac{dy}{dx} = 3y \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{3y - 3x^2}{3y^2 - 3x} = \frac{y - x^2}{y^2 - x}$$



Ex 3: Find an eqn. of the tangent line to the Folium of Descartes at $(\frac{3}{2}, \frac{3}{2})$.
At what point(s) in the first quadrant is the tangent line horizontal?

Sol: $m_{\text{tan}} = \frac{dy}{dx} \Big|_{(\frac{3}{2}, \frac{3}{2})}$

$$x^3 + y^3 = 3xy$$

$$\Rightarrow 3x^2 + 3y^2 \frac{dy}{dx} = 3(y + x \frac{dy}{dx})$$

$$\Rightarrow (y^2 - x) \frac{dy}{dx} = y - x^2$$

$$\Rightarrow \frac{dy}{dx} = \frac{y - x^2}{y^2 - x}$$

$$\therefore m_{\text{tan}} = \frac{dy}{dx} \Big|_{(\frac{3}{2}, \frac{3}{2})} = \frac{\frac{3}{2} - (\frac{3}{2})^2}{(\frac{3}{2})^2 - \frac{3}{2}} = -1$$

$\therefore$ the eqn. of the tangent line is $y - \frac{3}{2} = -1(x - \frac{3}{2})$
 $\hookrightarrow x + y = 3$

The tangent line is horizontal at the points

$$\text{Here, } \frac{y - x^2}{y^2 - x} = 0 \quad \left| \quad \begin{array}{l} x^3 + y^3 = 3xy \\ \Rightarrow x^3 = 2 \\ \Rightarrow x = \sqrt[3]{2}, \sqrt[3]{2} \cdot 0, \sqrt[3]{2} \cdot 2 \\ \Rightarrow x = 0, \sqrt[3]{2} \end{array} \right.$$

$$\therefore x^3 + y^3 = 3xy,$$

$$y^3 = 0; y = 0$$

$$\therefore 2 + y^3 = 3 \cdot \sqrt[3]{2} \cdot y$$

$$y = x^2 = (\sqrt[3]{2})^2$$

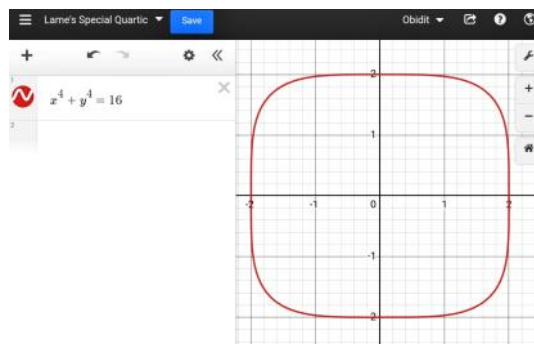
$$\therefore (x, y) = (0, 0), (\sqrt[3]{2}, (\sqrt[3]{2})^2)$$

$$\therefore (0, 0) \quad x \rightarrow \text{slope undefined}$$

$$(\sqrt[3]{2}, (\sqrt[3]{2})^2) \rightarrow \text{slope defined.}$$

⊗ Lamé's Special Quartic :-

$$x^4 + y^4 = 16$$



→ Find the slope of the tangent line at $(1, \sqrt[4]{15})$.

Solⁿ:

$$x^4 + y^4 = 16$$

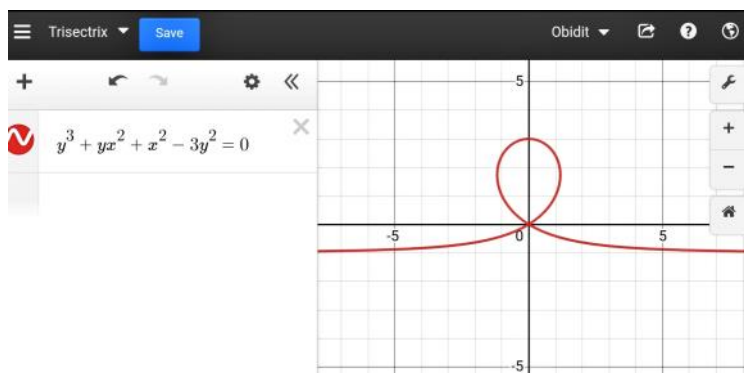
$$\Rightarrow 4y^3 \frac{dy}{dx} = -4x^3$$

$$\Rightarrow \frac{dy}{dx} = -\frac{x^3}{y^3}$$

$$\therefore \frac{dy}{dx} \Big|_{(1, \sqrt[4]{15})} = -\frac{1^3}{(\sqrt[4]{15})^3} = -0.1312$$

⊗ Trisectrix -

$$y^3 + yx^2 + x^2 - 3y^2 = 0$$



→ Find the slope of tangent line at $(0, 3)$.

Solⁿ:

$$y^3 + yx^2 + x^2 - 3y^2 = 0$$

Solⁿ:

$$y^3 + yx^2 + x^2 - 3y^2 = 0$$

$$\Rightarrow 3y^2 + x^2 \frac{dy}{dx} + 2xy + 2x - 6y \frac{dy}{dx} = 0$$

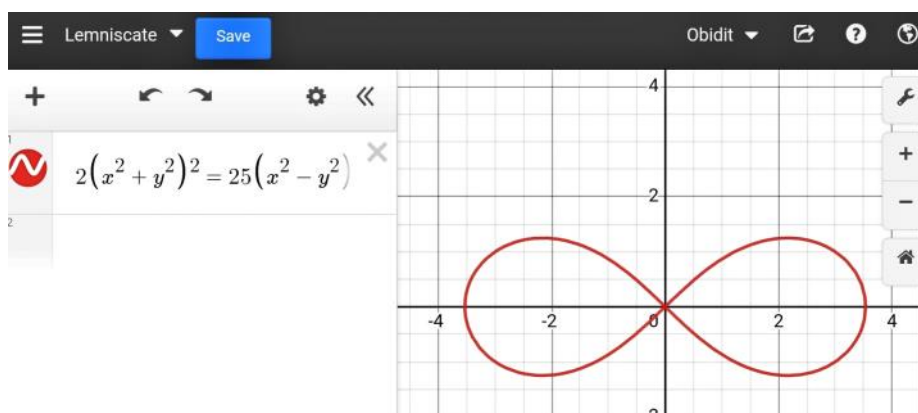
$$\Rightarrow \frac{dy}{dx} (Cy - x^2) = 3y^2 + 2xy + 2x$$

$$\Rightarrow \frac{dy}{dx} = \frac{3y^2 + 2xy + 2x}{Cy - x^2}$$

$$\therefore \left. \frac{dy}{dx} \right|_{(0,3)} = \frac{3 \cdot 3^2 + 2 \cdot 0 \cdot 3 + 2 \cdot 0}{6 \cdot 3 - 0} = \frac{27}{18} = \frac{3}{2}$$

⊗ Lemniscate -

$$2(x^2 + y^2)^2 = 25(x^2 - y^2)$$



Find the slope of the tangent line at point (3,1).

Solⁿ:

$$2(x^2 + y^2)^2 = 25(x^2 - y^2)$$

$$\Rightarrow 4(x^2 + y^2) \left(2x + 2y \frac{dy}{dx} \right) = 50 \left(x - \frac{dy}{dx} \right)$$

$$\Rightarrow 8 \left(x^3 + xy^2 \frac{dy}{dx} + y^2 \frac{dy}{dx} \right) = 50 \left(x - \frac{dy}{dx} \right)$$

$$\Rightarrow 8x^3 + 8xy^2 \frac{dy}{dx} + 8y^2 \frac{dy}{dx} = 50x - 50 \frac{dy}{dx}$$

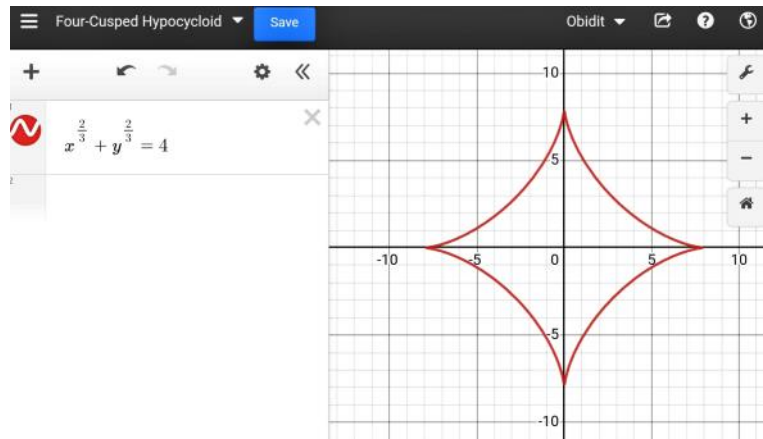
$$\Rightarrow 8xy^2 \frac{dy}{dx} - 50 \frac{dy}{dx} + 8y^2 \frac{dy}{dx} = 50x - 8x^3 - 8xy^2$$

$$\Rightarrow \frac{dy}{dx} = \frac{50x - 8x^3 - 8xy^2}{8xy^2 - 50 + 8y^2}$$

$$\therefore \left. \frac{dy}{dx} \right|_{(3,1)} = \frac{50 \cdot 3 - 8 \cdot 3^3 - 8 \cdot 3 \cdot 1^2}{8 \cdot 3 \cdot 1^2 - 50 + 8 \cdot 1^2} = -3$$

⊗ Four - Cusped Hypocycloid -

$$x^{2/3} + y^{2/3} = 4$$



~~Find~~ Find the slope of the tangent line at $(-1, 3\sqrt{3})$.

Soln:

$$x^{2/3} + y^{2/3} = 4$$

$$\Rightarrow \frac{2}{3}x^{-1/3} + \frac{2}{3}y^{-1/3} \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{x^{1/3}}{y^{1/3}} = -\frac{y^{1/3}}{x^{1/3}}$$

$$\therefore \left. \frac{dy}{dx} \right|_{(-1, 3\sqrt{3})} = -\frac{(3\sqrt{3})^{1/3}}{(-1)^{1/3}} = \sqrt{3}.$$

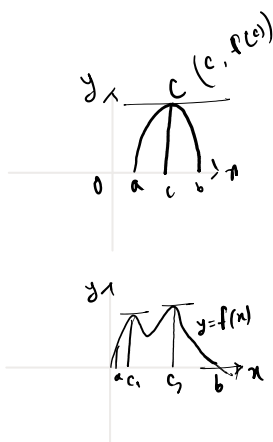
Expansion of functions

Rolle's Theorem:

Statement:

Let a function f be continuous on $[a, b]$, differentiable on (a, b) . If $f(a) = 0$ and $f(b) = 0$, then there exists a point $c \in (a, b)$ such that $f'(c) = 0$. Geometrically, if the graph of a differentiable function intersects the x -axis at $x = a$ and $x = b$, then we can find at least one c in (a, b) where the tangent line to the graph of $y = f(x)$ is horizontal.

(More in Anton's Book)



Taylor's Theorem:

Taylor series or Taylor expansion of a function is an

infinite sum of terms that are expressed in terms of the function's derivatives at a single point.

The Taylor series of a real or complex-valued function $f(x)$ that is infinitely differentiable at a real or complex number a is the power series.

$$f(a) + \frac{f'(a)}{1!} (x-a) + \frac{f''(a)}{2!} (x-a)^2 + \frac{f'''(a)}{3!} (x-a)^3 + \dots$$

Ex:

$$\ln x = (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 \dots \text{ is valid in } (0, 2).$$

Here, $a=1$

$f(x) = \ln x$	$f(1) = 0$	$\left\{ \begin{array}{l} \text{at least} \\ 3 \text{ nonzero} \\ \text{terms should} \\ \text{have in the} \\ \text{series to get 4th} \end{array} \right.$
$f'(x) = \frac{1}{x}$	$f'(1) = 1$	
$f''(x) = -\frac{1}{x^2}$	$f''(1) = -2$	
$f'''(x) = \frac{2}{x^3}$	$f'''(1) = 2$	

$$f^{(4)}(x) = \frac{2}{x^3} = \frac{(-1)^2 (2!)}{x^3}$$

$$\therefore f^{(n)}(x) = \frac{(-1)^{n-1} (n-1)!}{x^n}$$

Maclaurin's Theorem:

A Maclaurin series is a power series that allows one to calculate an approximation of a function $f(x)$ for input values close to zero, given that one knows the values of the successive derivatives of the function at zero.

If $f(x), f'(x), f''(x), \dots, f^{(n-1)}(x)$ are continuous on $[a, b]$ and if $f^{(n)}(x)$ exists in (a, b) at $a=0$, from Taylor's Series.

$$f(x) = f(0) + (x-0)f'(0) + \frac{(x-0)^2}{2!} f''(0) + \frac{(x-0)^3}{3!} f'''(0) + \dots$$

Ex: $\ln(1+x) = x - \frac{x^2}{2!} + \frac{x^3}{3!} - \frac{x^4}{4!} + \dots$

$$f(x) = \ln(1+x) \quad f(0) = 0$$

$$f'(x) = \frac{1}{1+x} \quad f'(0) = 1$$

$$f''(x) = -\frac{1}{(1+x)^2} \quad f''(0) = -1$$

$$f'''(x) = \frac{2}{(1+x)^3} \quad f'''(0) = 2$$

$$\therefore f^{(n)}(x) = \frac{(-1)^{n-1} (n-1)!}{(1+x)^n}$$

$$\begin{aligned} \therefore \ln(1+x) &= 0 + x + \frac{x^2}{2!}(-1) + \frac{x^3}{3!}(2) + \dots \\ &= x - \frac{x^2}{2} + \frac{x^3}{3} - \dots \end{aligned}$$

Taylor's theorem with Lagrange's form of remainder:

From Taylor's Series:

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!} f''(a) + \frac{(x-a)^3}{3!} f'''(a) + \dots + \frac{(x-a)^{n-1}}{(n-1)!} f^{(n-1)}(a)$$

(i) $f(x), f'(x), \dots, f^{(n-1)}(x) \leftarrow$ continuous in $[a, b]$

(ii) $f^{(n)}(x)$ exists in (a, b) we can find a number $\theta \in (0, 1)$ such that $R_n = ?$

Rewrite (3) in the form as

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!} f''(x) + \frac{h^3}{3!} f'''(x) + \frac{h^n}{n!} f^{(n)}(x+\theta h), \quad 0 < \theta < 1$$

$$R_n = \frac{h^n}{n!} f^{(n)}(x+\theta h), \quad 0 < \theta < 1 \quad \text{is the Lagrange's form of remainder.}$$

⊗ The analytical proof is left as an exercise.

Taylor's theorem with Cauchy's form of remainder:

$$R_n = \frac{h^n (1-\theta)^{n-1}}{(n-1)!} f^{(n)}(x+\theta h), \quad 0 < \theta < 1$$

Maclaurin's Theorem with Lagrange's form of remainder:

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots + R_n$$

$$R_n = \frac{x^n}{n!} f^{(n)}(\theta x), \quad 0 < \theta < 1$$

Maclaurin's theorem with Cauchy's form of remainder:

$$R_n = \frac{x^n (1-\theta)^{n-1}}{(n-1)!} f^{(n)}(\theta x), \quad 0 < \theta < 1.$$

Problem: Apply Maclaurin's theorem with Lagrange's form of remainder to prove that:

$$\log(1+x) = x - \frac{x^2}{2!} + \frac{x^3}{3!} - \frac{x^4}{4!} + \dots + (-1)^{n-2} \frac{x^{n-1}}{(n-1)!} + (-1)^{n-1} \frac{x^n}{n(1+\theta x)^n}; \quad 0 < \theta < 1.$$

Soln:

$$f(x) = \log(1+x)$$

$$f'(x) = \frac{1}{1+x} \quad f'(0) = 1$$

$$f''(x) = \frac{(-1) \cdot 1!}{(1+x)^2} \quad f''(0) = -1$$

$$f'''(x) = \frac{(-1)^2 2!}{(1+x)^3} \quad f'''(0) = 2$$

$$f^{(n)}(x) = \frac{(-1)^{n-1} (n-1)!}{(1+x)^n}$$

$$f^{(n)}(\theta x) = \frac{(-1)^{n-1} (n-1)!}{(1+\theta x)^n}$$

Therefore, the Maclaurin's theorem with Lagrange's form of remainder is

$$f(x) = 0 + x \cdot 1 + \frac{x^2}{2!} (-1) + \frac{x^3}{3!} 2 + \frac{(-1)^{n-2} (n-2)! \cdot x^{n-1}}{(n-1)!}$$

$$+ \frac{(-1)^{n-1} (n-1)! x^n}{(1+\theta x)^n n!}$$

Ex:

$$(i) \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + (-1)^{n-1} \frac{x^{2n-1}}{(2n-1)!} + (-1)^n \frac{x^{2n}}{(2n)!} (\sin \theta x), \quad 0 < \theta < 1$$

$$(ii) \log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \dots - \frac{x^{n-1}}{n-1} - \frac{x^n}{n(1-\theta x)^n}, \quad 0 < \theta < 1$$

Taylor's theorem with Lagrange's form of remainder:

$$\log(x+h) = \log x + \frac{h}{x} - \frac{h^2}{2x^2} + \dots + (-1)^{n-1} \frac{h^n}{n(x+\theta h)^n}, \quad 0 < \theta < 1$$

$$f(x+h) = \log(x+h)$$

$$f(x+h) = \log(x+h) \quad \frac{1}{n(x+\theta h)^n}, 0 < \theta < 1$$

$$f(x+h) = \log(x+h)$$

$$h=0, f(x) = \log x$$

$$f'(x) = \frac{1}{x}$$

$$f''(x) = -\frac{1}{x^2}$$

$$f'''(x) = \frac{2}{x^3}$$

$$f^{(n)}(x) = \frac{(-1)^{n-1} (n-1)!}{x^n}$$

$$f^{(n)}(x+\theta h) = \frac{(-1)^{n-1} (n-1)!}{(x+\theta h)^n}$$

Apply MacLaurin's Theorem with Cauchy's form of remainder.

$$f(x) = e^{ax} \sin px, f(0) = 0$$

$$f'(x) = e^{ax} [p \cos px + a \sin px], f'(0) = p$$

$$f''(x) = e^{ax} [(a^2 - p^2) \sin px + 2ap \cos px], f''(0) = 2ap$$

$$f'''(x) = e^{ax} [(a^3 - 3ap^2) \sin px + (3a^2p - p^3) \cos px], f'''(0) = 3a^2p - p^3 = p(3a^2 - p^2)$$

$$f^{(n)}(x) = e^{ax} (a^2 + p^2)^{n/2} \sin \left(px + n \tan^{-1} \frac{p}{a} \right)$$

$$f^{(n)}(0) = e^{a \cdot 0} (a^2 + p^2)^{n/2} \left(\sin p \cdot 0 + n \tan^{-1} \frac{p}{a} \right)$$

$$\therefore e^{ax} \sin px = p^{-1} + \frac{x^2}{2!} (2ap) + \frac{x^3}{3!} p(3a^2 - p^2) + \frac{x^4}{(n-1)!} (1-\theta)^{n-1} e^{a\theta x} (a^2 + p^2)^{n/2} \sin \left(p\theta x + n \tan^{-1} \frac{p}{a} \right)$$

Apply MacLaurin's Theorem to establish -

$$1) \log(\sec x) = \frac{x^2}{2} + \frac{x^4}{12} + \frac{x^6}{45} + \dots$$

$$2) \log(\cos x) = \frac{x^2}{2!} - 2 \frac{x^4}{4!} + 16 \frac{x^6}{6!} - \dots$$

$$3) \sec x = 1 + \frac{x^2}{2!} + \frac{5x^4}{4!} + \dots$$

$$4) \cos^2 x = 1 - x^2 + \frac{x^4}{3} + \dots$$

$$5) \tan^{-1}(x) = \frac{x}{1} + \frac{x^3}{3} - \frac{x^5}{5} + \frac{x^7}{7} + \dots$$

$$6) \log(1+x) = \log 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{16} + \dots$$

$$5) \tan^{-1}(x) = \frac{x}{1} + \frac{x^3}{3} - \frac{x^5}{5} + \frac{x^7}{7} + \dots$$

$$6) \log(1+x) = \log 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{128} + \dots$$

$$7) \log(1+\sin x) = x - \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{12} + \dots$$

$$8) e^{\cos x} = 1 + x - \frac{x^2}{2!} - 2 \cdot \frac{x^4}{4!} - 2^2 \frac{x^6}{6!} + \dots$$

Successive Differentiation

Ex 01: $y = \log(ax+b)$. Find y_n .

Solⁿ: $y = \log(ax+b)$

$$\Rightarrow y_1 = \frac{1}{ax+b} \cdot a = a(ax+b)^{-1}$$

$$\Rightarrow y_2 = (-1)a \cdot a(ax+b)^{-2} = (-1)a^2(ax+b)^{-2}$$

$$\Rightarrow y_3 = (-1)^2 2! a^3(ax+b)^{-3}$$

...

$$y_n = (-1)^{n-1} (n-1)! a^n (ax+b)^{-n}.$$

Ex 02:

$$y = \sin(ax+b)$$

$$\Rightarrow y_1 = a \cos(ax+b) \\ = a \sin\left(\frac{\pi}{2} + ax+b\right)$$

$$\Rightarrow y_2 = a^2 \sin\left[\frac{\pi}{2} + \left(\frac{\pi}{2} + (ax+b)\right)\right] \\ = a^2 \left[2 \cdot \frac{\pi}{2} + ax+b\right]$$

$$\Rightarrow y_3 = a^3 \sin\left[3\frac{\pi}{2} + ax+b\right]$$

$$y_n = a^n \sin \left[n \cdot \frac{\pi}{2} + a n + b \right]$$

Ex 03:

$$y = \frac{2x+3}{x^2+3x+2}$$

$$\Rightarrow y = \frac{(2x+3)}{(x+1)(x+2)} = \frac{1}{x+1} + \frac{1}{x+2} = (x+1)^{-1} + (x+2)^{-1}$$

$$\Rightarrow y_1 = (-1)(x+1)^{-2} + (-1)(x+2)^{-2}$$

$$\Rightarrow y_2 = (-1)(-2) \left[(x+1)^{-3} + (x+2)^{-3} \right]$$

$$\Rightarrow y_2 = (-1)^2 2! \left[(x+1)^{-(2+1)} + (x+2)^{-(2+1)} \right]$$

$$y_n = (-1)^n n! \left[(x+1)^{-(n+1)} + (x+2)^{-(n+1)} \right]$$

Ex 04:

$$y = \cos 3x \cos 2x \cos x$$

$$= \frac{1}{2} (2 \cos 3x \cos 2x) \cdot \cos x$$

$$= \frac{1}{2} [\cos 5x + \cos x] \cos x$$

$$= \frac{1}{4} (2 \cos 5x \cos x) + \frac{1}{4} (2 \cos x)$$

Formula

$$2 \cos x \cos y = \cos(x+y) + \cos(x-y)$$

$$= \frac{1}{4} (2\cos 5x \cos x) + \frac{1}{4} (2\cos x) \quad |$$

$$= \frac{1}{4} (\cos 6x + \cos 4x + \cos 2x + 1)$$

$$y_1 = \frac{1}{4} [-6\sin 6x - 4\sin 4x - 2\sin 2x]$$

$$= \frac{1}{4} [6\cos(\pi/2 + 6x) + 4\cos(\pi/2 + 4x) + 2\cos(\pi/2 + 2x)]$$

$$y_2 = \frac{1}{4} [-6^2\sin(\pi/2 + 6x) - 4^2\sin(\pi/2 + 4x) - 2^2\sin(\pi/2 + 2x)]$$

$$= \frac{1}{4} [6^2\cos(2\pi/2 + 6x) + 4^2\cos(2\pi/2 + 4x) + 2^2\cos(2\pi/2 + 2x)]$$

$$y_n = \frac{1}{4} [6^n\cos(n\cdot\frac{\pi}{2} + 6x) + 4^n\cos(n\cdot\frac{\pi}{2} + 4x) + 2^n\cos(n\cdot\frac{\pi}{2} + 2x)]$$

Complex Number Set

$$\hookrightarrow \mathbb{C} = \{(x, y) = x + iy : x, y \in \mathbb{R}, i = \sqrt{-1}\}$$

$$\left. \begin{aligned} e^{ix} &= \cos x + i\sin x \\ e^{-ix} &= \cos x - i\sin x \end{aligned} \right\} \text{Euler's identity.}$$

$$(\cos x + i\sin x)^n = \cos(nx) + i\sin(nx) = e^{i(nx)}$$

→ De Moivre's identity.

$$(\cos x - i\sin x)^n = e^{-i(nx)}$$

$$z \in \mathbb{C}$$

$$z = x + iy$$

$$z \in \mathbb{C}$$

$$z = x + iy$$

$$\operatorname{Re}(z) = \text{Real part of } z = x$$

$$\operatorname{Im}(z) = \text{Imaginary part of } z = y$$

$$z = \cos x + i \sin x$$

$$z^{-1} = \frac{1}{z} = \cos x - i \sin x$$

$$z + \frac{1}{z} = 2 \cos x$$

$$\Rightarrow \left(z + \frac{1}{z}\right)^n = 2^n \cos^n x$$

$$\Rightarrow \left(z + \frac{1}{z}\right)^n = z^n + {}^nC_1 z^{n-1} \frac{1}{z} + {}^nC_2 z^{n-2} \frac{1}{z^2} + {}^nC_3 z^{n-3} \frac{1}{z^3} + {}^nC_4 z^{n-4} \frac{1}{z^4} + \dots + {}^nC_n \frac{1}{z^n}$$

$${}^nC_n = \binom{n}{n}$$

$${}^nC_1 = n$$

$${}^nC_2 = \frac{n(n-1)}{2}$$

$$\therefore {}^nC_n = {}^nC_{n-n}$$

$$\Rightarrow \left(z + \frac{1}{z}\right)^5 = z^5 + 5z^3 + 10z + 10\frac{1}{z} + 5\frac{1}{z^3} + \frac{1}{z^5}$$

$$z^n = \cos(nx) + i \sin(nx)$$

$$z^{-n} = \frac{1}{z^n} = \cos(nx) - i \sin(nx)$$

$$z^n + \frac{1}{z^n} = 2 \cos(nx)$$

$$z^n - \frac{1}{z^n} = 2i \sin(nx)$$

Ex 07:

$$\text{Given, } y = \cos^5 x$$

$$\text{Let } z = \cos x + i \sin x$$

$$z^{-1} = \frac{1}{z} = \cos x - i \sin x$$

$$z + \frac{1}{z} = 2 \cos x$$

$$z = 2e^{i\pi/2}$$

$$\Rightarrow \left(2 + \frac{1}{z}\right) = 2\cos x$$

$$\Rightarrow \left(2 + \frac{1}{z}\right)^5 = 2^5 \cos^5 x$$

$$\Rightarrow \cos^5 x = y = \frac{1}{2^5} \left(2 + \frac{1}{z}\right)^5$$

$$\Rightarrow y = \frac{1}{2^5} [2\cos^5 x + 5(2\cos^3 x) + 10(2\cos x)]$$

$$\Rightarrow y = \frac{1}{2^5} [\cos^5 x + 5\cos^3 x + 10\cos x]$$

$$\Rightarrow y_1 = \frac{1}{2^5} [-5\sin^5 x - 5 \times 3\sin^3 x - 10\sin x]$$

$$\Rightarrow y_1 = \frac{1}{2^5} [5\cos(\pi/2 + 5x) + 5 \times 3\cos(\pi/2 + 3x) + 10\cos(\pi/2 + x)]$$

$$\Rightarrow y_2 = \frac{1}{2^5} [-5^2 \sin(\pi/2 + 5x) - 5 \times 3^2 \sin(\pi/2 + 3x) - 10 \sin(\pi/2 + x)]$$

⋮

$$y_n = \frac{1}{2^5} [5^n \cos(n\frac{\pi}{2} + 5x) + 5 \times 3^n \cos(n\frac{\pi}{2} + 3x) + 10 \cos(n\frac{\pi}{2} + x)]$$

Ex 06: $y = \sin^6 x$

$$\text{Here, } z - \frac{1}{z} = 2i \sin x$$

$$\Rightarrow \left(z - \frac{1}{z}\right)^6 = 2^6 \cdot i^6 \cdot \sin^6 x$$

$$\left(z - \frac{1}{z} \right)^6 = z^6 - 6z^5 \cdot \frac{1}{z} + 15z^4 \cdot \frac{1}{z^2} - 20z^3 \cdot \frac{1}{z^3} + 15z^2 \cdot \frac{1}{z^4} - 6z \cdot \frac{1}{z^5} + \frac{1}{z^6}$$

$$= z^6 - 6z^4 + 15z^2 - 20 + 15 \frac{1}{z^2} - 6 \frac{1}{z^4} + \frac{1}{z^6}$$

$$\Rightarrow y = \sin^6 x = -\frac{1}{2^6} [2\cos 6x - 6 \times 2\cos 4x + 15 \times 2\cos 2x - 20]$$

$$\Rightarrow y = \sin^6 x = -\frac{1}{2^5} [2\cos 6x - C \times 2\cos 4x + 14 \times 2\cos 2x - 20]$$

$$= -\frac{1}{2^5} [\cos 6x - C\cos 4x + 14\cos 2x - 10]$$

⋮

$$J_n = -\frac{1}{2^n} [C^n \cos(n\frac{\pi}{2} + 6x) - C \times 4^n \cos(n\frac{\pi}{2} + 4x) + 14 \times 2^n \cos(n\frac{\pi}{2} + 2x)]$$

Liebniz Theorem

If $u(x)$ and $v(x)$ are functions of x , then -

$$(uv)_n = u_n v + {}^n C_1 u_{n-1} v_1 + {}^n C_2 u_{n-2} v_2 + \dots + {}^n C_n u_{n-n} v_n + \dots + u v_n$$

where the suffixes of u and v denote differentiation of different orders with respect to x .

Solved Problems 1 and 2

Sunday, October 29, 2023 7:11 PM

Chapter 1: Problems on Limits, Continuity and Differentiability

1. A function $f(x)$ is defined as follows:

$$f(x) = \begin{cases} 6+4x, & -\frac{3}{2} \leq x < 0 \\ 6-4x, & 0 \leq x < \frac{3}{2} \\ 6-4x, & x \geq \frac{3}{2} \end{cases}$$

Discuss the continuity at $x = \frac{3}{2}$ and differentiability at $x = 0$. Also sketch the graph of $f(x)$.

2. A function $f(x)$ is defined by

$$f(x) = \begin{cases} 2x-1, & x \leq 1 \\ x^2-x+1, & x > 1 \end{cases}$$

Examine the continuity and differentiability of $f(x)$ at $x = 1$. Also sketch the graph of $f(x)$.

3. A function $f(x)$ is defined by

$$f(x) = \begin{cases} 1+x, & x \leq 0 \\ x, & 0 < x < 1 \\ 2-x, & 1 \leq x \leq 2 \\ 2x-x^2, & x > 2 \end{cases}$$

Examine the continuity and differentiability of $f(x)$ at $x = 1$ and $x = 2$. Also sketch the graph of $f(x)$.

4. A function $f(x)$ is defined by

$$f(x) = \begin{cases} \frac{x-3}{x-3}, & x \neq 3 \\ 0, & x = 3 \end{cases}$$

Find $\lim_{x \rightarrow 3} f(x)$, if it exists.

5. A function $f(x)$ is defined by

$$f(x) = \begin{cases} x, & 0 \leq x < 1 \\ 2-x, & 1 \leq x \leq 2 \\ \frac{1}{2}x^2, & x > 2 \end{cases}$$

Examine the continuity and differentiability of $f(x)$ at $x = 2$. Also sketch the graph of $f(x)$.

6. A function $f(x)$ is defined by

6. A function $f(x)$ is defined by

$$f(x) = \begin{cases} x^2+1, & 0 < x \leq 1 \\ 4x^2-3x, & 1 < x < 2 \\ 3x+4, & x \geq 2 \end{cases}$$

Discuss the continuity of $f(x)$ at $x = 1$ and the existence $f'(x)$ at $x = 2$. Also sketch the graph of $f(x)$.

7. A function $f(x)$ is defined by

$$f(x) = \begin{cases} x^2+1, & x \leq 0 \\ 2x-1, & 0 < x \leq 1 \\ x^2-x+1, & x > 1 \end{cases}$$

Test the continuity and differentiability of $f(x)$ at $x = 0$ and at $x = 1$. Also sketch the graph of $f(x)$.

8. A function $f(x)$ is defined by

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x > 0 \\ -x^2, & x \leq 0 \end{cases}$$

Discuss the continuity and differentiability of $f(x)$ at $x = 0$.

9. Consider the function

$$f(x) = \begin{cases} 2x+5, & -3 \leq x < 0 \\ -x, & x = 0 \\ -5x^2, & x > 0 \end{cases}$$

Discuss the continuity and differentiability of $f(x)$ at $x = 0$.

10. Consider the function

$$f(x) = \begin{cases} x^2, & x \leq 0 \\ x, & 0 < x < 1 \\ \frac{1}{x}, & x \geq 1 \end{cases}$$

Discuss the continuity and differentiability of $f(x)$ at $x = 0$ and $x = 1$.

① Sol^y

Here, $f(x) = \begin{cases} 6+4x & -\frac{3}{2} \leq x < 0 \\ 6-4x & 0 \leq x < \frac{3}{2} \\ 6-4x & x \geq \frac{3}{2} \end{cases}$ Here, at $x = \frac{3}{2}$, $f(x)$ is defined.

if, $x = \frac{3}{2}$, $\lim_{x \rightarrow \frac{3}{2}^-} f(x) = 6-4(\frac{3}{2}) = 0$

$\lim_{x \rightarrow \frac{3}{2}^+} f(x) = 6+4(\frac{3}{2}) = 12$

Here, L.H.L $\neq$ R.H.L, Thus $f(x)$ is not continuous at $x = \frac{3}{2}$.

Then, at $x = 0$, $\lim_{x \rightarrow 0^-} f(x) = 6+4(0) = 6$

$\lim_{x \rightarrow 0^+} f(x) = 6-4(0) = 6$

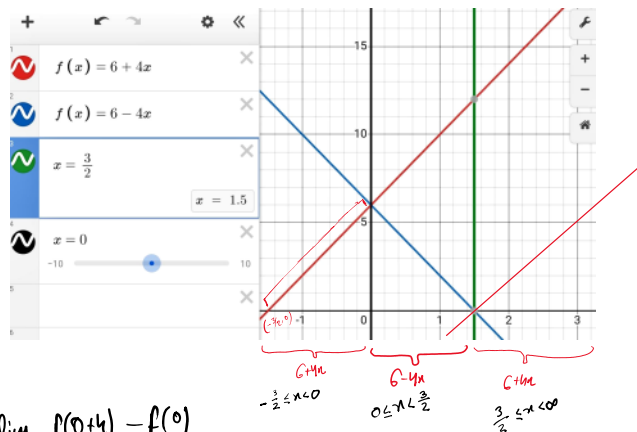
Here, L.H.L = R.H.L, Thus $f(x)$ is continuous at $x = 0$.

So, $f(x)$ at $x = 0$ might be differentiable.

Now, $f'_L(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$
 $= \lim_{h \rightarrow 0} \frac{6+4(0+h) - 6-4(0)}{h}$
 $= \lim_{h \rightarrow 0} \frac{6+4h-6}{h}$
 $= 4$

Now, $f'_R(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$
 $= \lim_{h \rightarrow 0} \frac{6-4(0+h) - 6-4(0)}{h}$
 $= \lim_{h \rightarrow 0} \frac{6-4h-6}{h}$
 $= -4$

$\therefore f'_R(0) \neq f'_L(0) \Rightarrow$ Hence, the function is not differentiable at $x = 0$.



② Solⁿ:

Hence, $f(x) = \begin{cases} 2x-1, & x \leq 1 \\ x^2-x+1, & x > 1 \end{cases}$ Here, at $x=1$, $f(x)$ is defined

$$\therefore x=1, \lim_{x \rightarrow 1^-} f(x) = 2(1)-1 = 1$$

$$\lim_{x \rightarrow 1^+} f(x) = 1^2-1+1 = 1$$

Hence, L.H.L = R.H.L, Thus $f(x)$ is continuous at $x=1$.

$$\begin{aligned} \text{Now, } f'_L(1) &= \frac{f(1+h) - f(1)}{h} \\ &= \frac{2(1+h)-1 - 2(1)+1}{h} \\ &= \frac{2+2h-1-2+1}{h} \\ &= 2 \end{aligned}$$

$$\begin{aligned} \text{Now, } f'_R(1) &= \frac{f(1+h) - f(1)}{h} \\ &= \frac{(1+h)^2 - (1+h) + 1 - (1^2-1+1)}{h} \\ &= \frac{1+2h+h^2 - (1+h+1-1+1)-1}{h} \\ &= \frac{h^2+h}{h} \\ &= h+1 \\ &= 1 \quad (h \rightarrow 0) \end{aligned}$$

Hence, $x \rightarrow 1$

Assuming, $x = h+1$

So, if $x \rightarrow 1$ then, $h \rightarrow 0$

$\therefore f'_R(1) \neq f'_L(1) \Rightarrow$ Hence the function is not differentiable at $x=1$.

③ Solⁿ:

Hence, $f(x) = \begin{cases} 1+x, & x \leq 0 \\ x, & 0 < x < 1 \\ 2-x, & 1 \leq x \leq 2 \\ 2x-x^2, & x > 2 \end{cases}$ Here, at $x=1$ & $x=2$, $f(x)$ is defined.

$$\therefore x=1, \lim_{x \rightarrow 1^-} f(x) = 1$$

$$\lim_{x \rightarrow 1^+} f(x) = 2-1 = 1$$

Hence, L.H.L = R.H.L, Thus $f(x)$ is continuous at $x=1$

$$\therefore x=2, \lim_{x \rightarrow 2^-} f(x) = 2-2 = 0$$

$$\lim_{x \rightarrow 2^+} f(x) = 2 \cdot 2 - 2^2 = 0$$

Hence, L.H.L = R.H.L, Thus $f(x)$ is continuous at $x=2$.

For, $x=1$,

$$f'_L(1) = \frac{f(1+h) - f(1)}{h} \quad \left| \quad f'_R(1) = \frac{f(1+h) - f(1)}{h} \right.$$

$$f'_L(1) = \frac{f(1+h) - f(1)}{h} \quad \left| \quad f'_R(1) = \frac{f(1+h) - f(1)}{h} \right.$$

$$= \frac{1+h-1}{h} \quad \left| \quad = \frac{2-1-h-2+h}{h} \right.$$

$$= 1 \quad \left| \quad = -1 \right.$$

$$\therefore f'_L(1) \neq f'_R(1)$$

$\therefore$ At $x=1$, $f(x)$ is continuous but not differentiable.

for $x=2$,

$$f'_L(2) = \frac{f(2+h) - f(2)}{h} \quad \left| \quad f'_R(2) = \frac{f(2+h) - f(2)}{h} \right. \quad \left. \begin{array}{l} \text{Here, } x \rightarrow 2 \\ \text{Assuming, } x = h+2, \\ \text{if } x \rightarrow 2 \text{ then } h \rightarrow 0. \end{array} \right.$$

$$= \frac{2 - (2+h) - 2+2}{h} \quad \left| \quad = \frac{2(2+h) - (2+h)^2 - 2 \cdot 2 + 2^2}{h} \right.$$

$$= -1 \quad \left| \quad = \frac{4+2h-4-h^2-h^2}{h} \right.$$

$$\quad \left| \quad = \frac{h^2-2h}{h} \right.$$

$$\quad \left| \quad = h-2 = -2 \quad (h \rightarrow 0)$$

$$\therefore f'_L \neq f'_R$$

$\therefore$ At $x=2$, $f(x)$ is continuous but not differentiable.

[The rest of the maths fall in the same type.
you can solve them on your own. You
can double check the answers from here. I collected
the answers from a topper 😊]

4 $\rightarrow$ limit does not exist at $x=3$.

5 $\rightarrow$ both continuous and differentiable at $x=2$.

6 $\rightarrow$ not continuous at $x=1$ and not differentiable $x=2$.

7 $\rightarrow$ limit does not exist at $x=0$, thus not continuous at $x=0$
continuous at $x=1$; not differentiable at $x=1$.

8 $\rightarrow$ having a bit confusion here. Probably not continuous
as it is not defined at $x=0$.

9 $\rightarrow$ limit does not exist

10 $\rightarrow$ continuous but not differentiable at $x=0$.

also continuous but not differentiable at $x=1$.

Differentiation Explicit and Implicit Functions

1. If $f(x) = \ln \frac{\sqrt{a+bx} - \sqrt{a-bx}}{\sqrt{a+bx} + \sqrt{a-bx}}$, find for what values of x , $\frac{1}{f'(x)} = 0$.

2. Find $\frac{dy}{dx}$, where $y = \frac{(x^3 - 2x + 5)^4 \sqrt{x^2 + 3x + 2}}{x^2 \tan \frac{1}{2}x}$.

3. If $f(x) = \left(\frac{a+x}{b+x}\right)^{a+b+2x}$, show that $f'(0) = \left(2 \log \frac{a}{b} + \frac{b^2 - a^2}{ab}\right) \left(\frac{a}{b}\right)^{a+b}$.

① Solⁿ: $f(x) = \ln \frac{\sqrt{a+bx} - \sqrt{a-bx}}{\sqrt{a+bx} + \sqrt{a-bx}} = y$

$$\Rightarrow y = \ln(\sqrt{a+bx} - \sqrt{a-bx}) - \ln(\sqrt{a+bx} + \sqrt{a-bx})$$

$$\Rightarrow y_1 = \frac{1}{\sqrt{a+bx} - \sqrt{a-bx}} \left(\frac{b}{2\sqrt{a+bx}} + \frac{b}{2\sqrt{a-bx}} \right) - \frac{1}{\sqrt{a+bx} + \sqrt{a-bx}} \left(\frac{b}{2\sqrt{a+bx}} - \frac{b}{2\sqrt{a-bx}} \right)$$

$$\Rightarrow y_1 = \frac{b}{2} \frac{1}{\sqrt{a+bx} - \sqrt{a-bx}} \left(\frac{\sqrt{a-bx} + \sqrt{a+bx}}{\sqrt{a+bx}\sqrt{a-bx}} \right) - \frac{b}{2} \left(\frac{1}{\sqrt{a+bx} + \sqrt{a-bx}} \right) \frac{\sqrt{a+bx} - \sqrt{a-bx}}{\sqrt{(a+bx)(a-bx)}}$$

$$\Rightarrow y_1 = \frac{b}{2\sqrt{a+bx}\sqrt{a-bx}} \left(\frac{\sqrt{a-bx} + \sqrt{a+bx}}{\sqrt{a+bx} - \sqrt{a-bx}} - \frac{\sqrt{a-bx} - \sqrt{a+bx}}{\sqrt{a+bx} + \sqrt{a-bx}} \right)$$

$$\Rightarrow y_1 = \frac{b}{2\sqrt{a+bx}\sqrt{a-bx}} \left(\frac{4\sqrt{a-bx}\sqrt{a+bx}}{a+bx - a-bx} \right)$$

$$\Rightarrow y_1 = \frac{2b}{2bx} = \frac{1}{x}$$

$$\therefore \frac{1}{f'(x)} = \frac{1}{1/x} = x = 0.$$

② Solⁿ: $y = \frac{(x^3 - 2x + 5)^4 \sqrt[3]{x^2 + 3x + 2}}{x^2 \tan \frac{1}{2}x}$

$$\Rightarrow \ln y = 4 \ln(x^3 - 2x + 5) + \frac{1}{3} \ln(x^2 + 3x + 2) - \ln(x^2 \tan \frac{1}{2}x)$$

$$\Rightarrow \frac{1}{y} y_1 = 4 \frac{1}{x^3 - 2x + 5} (3x^2 - 2) + \frac{1}{3} \frac{1}{x^2 + 3x + 2} (2x + 3) - \frac{1}{x^2 \tan \frac{1}{2}x} \left(\frac{x^2}{2} \sec^2 \frac{1}{2}x - 2x \tan \frac{1}{2}x \right)$$

$$\Rightarrow \frac{dy}{dx} = \left\{ \frac{4(3x^2 - 2)}{x^3 - 2x + 5} + \frac{2x + 3}{3(x^2 + 3x + 2)} - \frac{x^2/2 \sec^2 \frac{1}{2}x - 2x \tan \frac{1}{2}x}{x^2 \tan \frac{1}{2}x} \right\} \frac{(x^3 - 2x + 5)^4 \sqrt[3]{x^2 + 3x + 2}}{x^2 \tan \frac{1}{2}x}$$

③ Solⁿ: $y = \left(\frac{a+x}{b+x}\right)^{a+b+2x}$

$$\Rightarrow \ln y = (a+b+2x) \ln \left(\frac{a+x}{b+x}\right)$$

$$\Rightarrow \frac{1}{j} \delta_1 = (a+b+2n) \left(\frac{b+n}{a+n} \right) \frac{(b+n)-(a+n)}{(b+n)^2} + 2 \ln \left(\frac{a+n}{b+n} \right)$$

$$\Rightarrow \frac{1}{j} \delta_1 = \frac{(a+b+2n)(b-a)}{(a+n)(b+n)} + 2 \ln \left(\frac{a+n}{b+n} \right)$$

$$\Rightarrow f'(x) = \left\{ \frac{(a+b+2n)(b-a)}{(a+n)(b+n)} + 2 \ln \left(\frac{a+n}{b+n} \right) \right\} \left(\frac{a+n}{b+n} \right)^{a+b+2n}$$

$$\Rightarrow f'(0) = \left\{ \frac{(a+b)(b-a)}{ab} + 2 \ln \left(\frac{a}{b} \right) \right\} \left(\frac{a}{b} \right)^{a+b}$$

$$\Rightarrow f'(0) = \left(2 \log \frac{a}{b} + \frac{b^2 - a^2}{ab} \right) \left(\frac{a}{b} \right)^{a+b}$$

[showed]